UNIVERSITY
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Master's thesis

Topological quantum computing with the Kitaev chain

Solving quantum computing and
providing the world with infinite porridge

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Abstract

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Chapter 2

The Kitaev chain

In this chapter we closely follow the original theory laid out by Kitaev [1], re-deriving the main results and filling in with intermediate steps and discussion for clarity. The Kitaev chain is a toy model of a one dimensional spinless p-wave superconductor, and is an example of material that can realize a topological superconducting phase. It consists of a chain of spinless fermions with nearest-neighbour hopping and superconducting pairing, under the influence of a chemical potential. The Hamiltonian takes the form

$$H = \sum_j \left[-\omega \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) - \mu \left(a_j^\dagger a_j - \frac{1}{2} \right) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right] \tag{2.1}$$

where a_j and $a_j^\dagger$ are annihilation and creation operators for the fermionic sites j in the chain. ω is a hopping amplitude, μ is the chemical potential, and $\Delta = |\Delta| e^{i\theta}$ is the induced superconducting gap. The Hamiltonian serves to give rise to Majorana fermion quasiparticles at its boundaries. These Majorana fermions can be used as qubits, and a network of chains can be used for quantum computation.

Majorana fermions are interesting candidates for qubits as their topological nature provides robustness against local perturbations and certain sources of decoherence. Their non-Abelian nature also allows for braiding, where the majorana fermions exchange position and information about the path is stored in the wavefunction.

2.1 Majorana fermions

A Majorana fermion is a particle that is its own antiparticle. The Majorana operators can be defined as

$$c_{2j-1} = a_j + a_j^\dagger, \quad c_{2j} = \frac{a_j - a_j^\dagger}{i}, \quad (j = 1, ..., N) \tag{2.2}$$

which are hermitian, and anticommute, meaning they satisfy the relations

$$c_m^\dagger = c_m, \quad \{c_l, c_m\} = c_l c_m + c_m c_l = 2\delta_{lm}, \quad (l, m = 1, ..., 2N). \tag{2.3}$$

Using the definitions of the majorana operators 2.2 we can find the number operator for a site j

$$a_j^\dagger a_j = \frac{1}{2} \left(1 + i c_{2j-1} c_{2j} \right), \tag{2.4}$$

where the term $ic_{2j-1}c_{2j}$ corresponds to a phase error. If c_{2j-1} and c_{2j} are not on the same site, but separated in space, then $c_{2j-1}c_{2j}$ would represent an interaction between two separated Majorana sites, requiring the environment to couple the Majorana fermions simultaneously to create the phase error. Physical noise and imperfections are typically local, so placing the Majorana sites far enough away from each other would decrease the risk of a phase error. Further, a single Majorana operator cannot appear by itself in a reasonable Hamiltonian as it would not preserve fermionic parity; flipping the system from an even to odd (or vice versa) number of fermions. As such, the Majorana operators will always come in pairs $c_i c_l$, and thus no parity preserving term can involve just one isolated Majorana. Together this tells us that errors would either have to break parity, or couple both separated Majoranas at once, which is highly unlikely and thus the Majorana fermions are protected from local errors.

2.1.1 Rewriting the Hamiltonian in terms of Majorana operators

We can express the Hamiltonian in terms of Majorana operators. As we will see, this transformation will provide insight about the structure of the phases and the emergence of the edge modes. Before doing so we simplify the calculation by hiding the phase θ in the majorana operators

$$c_{2j-1} = e^{i\frac{\theta}{2}}a_j + e^{-i\frac{\theta}{2}}a_j^\dagger \tag{2.5}$$

$$c_{2j} = -ie^{i\frac{\theta}{2}}a_j + ie^{-i\frac{\theta}{2}}a_j^\dagger. \tag{2.6}$$

We can now perform the transformation of the Hamiltonian by inverting these relations and inserting into the original Hamiltonian 2.1. Together with the hermiticity and anticommutation relations 2.3 we get the Hamiltonian

$$H = \frac{i}{2}\sum_j \left[-\mu c_{2j-1}c_{2j} + (\omega + |\Delta|)\, c_{2j}c_{2j+1} + (-\omega + |\Delta|)\, c_{2j-1}c_{2j+2}\right]. \tag{2.7}$$

This calculation is shown in appendix A.1. Staring at this Hamiltonian, we can first see that it is on the general quadratic Majorana form

$$H = \frac{i}{4}\sum_{l,m} A_{lm}c_l c_m, \quad A^T = -A \tag{2.8}$$

with A being a real and antisymmetric matrix, encoding the intrinsic particle-hole symmetry of the superconducting system. The single particle spectrum is fully determined by the eigenvalues of A, reducing the problem from a 2^L-dimensional many-body operator problem in Fock space to a single-particle linear algebra problem on a $2L \times 2L$ matrix. This is the basis of BdG, as we will get back to later. Further, we see that all direct mentions of the complex phase θ are gone from H. This reflects that θ can be removed by an appropriate gauge transformation, and is therefore irrelevant for the bulk spectrum and the topological classification of the phases, which we will also touch upon later.

The Majorana representation also gives us a geometric picture of the chain. The Hamiltonian now consists of three terms, providing three different types of couplings. An on-site term $-\mu c_{2j-1}c_{2j}$, and two nearest-neighbour terms $(\omega + |\Delta|)c_{2j}c_{2j+1}$ and $(-\omega + |\Delta|)c_{2j-1}c_{2j+2}$. The on-site term couples two Majorana modes per site j, so for a chain of length L we get $2L$ Majorana operators. We can garner further insight by studying specific cases for the parameters μ, ω and $|\Delta|$ of the Hamiltonian.

Trivial case $\omega = |\Delta| = 0, \mu > 0$

In the case where μ dominates the Hamiltonian reduces to

$$H = -\mu \sum_j \left(a_j^\dagger a_j - \frac{1}{2} \right) = \frac{i}{2} \mu \sum_j c_{2j-1} c_{2j}. \tag{2.9}$$

In this case the system is topologically trivial as the Majoranas on each site are glued together to form a complex fermion. There are no unpaired Majorana fermions at the edges of the chain, making this case uninteresting for further study.

Topological case $\omega = |\Delta| > 0, \mu = 0$

In the case where the hopping and superconducting parameters dominate the Hamiltonian reduces to

$$H = i \sum_j \omega c_{2j} c_{2j+1}, \tag{2.10}$$

where the nearest neighbours between different sites dominate. This is an interesting case as c_2 couples to c_3, c_4 couples to c_5, and so on. But c_1 and c_{2L}, the Majorana edge states, do not appear in the Hamiltonian. They have however not disappeared physically, but rather remain as zero-energy Majorana modes localized at the edges of the wire. Together they form a nonlocal fermionic zero mode. This is the most important insight from the transformation of the Hamiltonian as these are the states we were looking to create.

We can now define new annihilation and creation operators spanning the sites j and $j+1$

$$d_j = \frac{1}{2} \left(c_{2j} + i c_{2j+1} \right) \tag{2.11}$$

$$d_j^\dagger = \frac{1}{2} \left(c_{2j} - i c_{2j+1} \right). \tag{2.12}$$

Such that we can rewrite the Hamiltonian

$$H = 2\omega \sum_{j=1} \left(d_j^\dagger d_j - \frac{1}{2} \right). \tag{2.13}$$

As the edge states are not mentioned in the Hamiltonian, this expression fully determines the energy spectrum of the bulk. We see that the bulk consists of fermionic modes with energy $\propto 2\omega$. Further, the ground states satisfy the condition $d_j|\psi\rangle = 0$ for all j. As such, all these modes will remain empty, and any ground state $|\psi\rangle$ is fully determined by the edge majoranas c_1 and c_{2L}. Therefore we get two orthogonal states where the above ground state condition for the bulk is true; $|\psi_0\rangle$ and $|\psi_1\rangle$. These ground states will be eigenstates of the hermitian operator $-ic_1c_{2L}$ as the edge Majoranas represent unpaired degrees of freedom. This operator also squares to 1, and as such has eigenvalues ± 1. We get

$$-ic_1c_{2L}|\psi_0\rangle = |\psi_0\rangle, \quad -ic_1c_{2L}|\psi_1\rangle = -|\psi_1\rangle, \tag{2.14}$$

where the labeling of the states were chosen arbitrarily. We see that the total fermion parity operator

$$P = \prod_j (-ic_{2j-1}c_{2j}) \tag{2.15}$$

has the same contribution from the bulk for both ground states. So the difference in parity is solely a result of the edge contribution $-ic_1c_{2L}$. The even parity of $|\psi_0\rangle$ represents the state being a superposition of states with an even number of electrons. And $|\psi_1\rangle$ has an odd parity, representing an odd number of electrons. Together they reflect the boundary degrees of freedom from the unpaired Majorana modes at the edges.

Arbitrary values of ω, μ, Δ

For arbitrary values of ω, μ, Δ the Hamiltonian has the general quadratic form (2.8) in terms of the Majorana operators. This means the Hamiltonian can be diagonalized into independent fermionic modes by a change of basis, implemented by a matrix W. We require that W is a $2N \times 2N$ real orthogonal ($W^TW = WW^T = I$) matrix, and that it has two further properties; the first being that it defines new operators $\{b'_m, b''_m\}$ by the linear transformation $b = Wc$, or

$$\begin{pmatrix} b'_1 \\ b''_1 \\ \vdots \\ b'_N \\ b''_N \end{pmatrix} = W \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_{2N-1} \\ c_{2N} \end{pmatrix}. \tag{2.16}$$

This property is ensured by W being real and orthogonal, such that after the transformation the Majorana anticommutation relations are preserved. This can be shown using $b_\gamma = \sum_\alpha W_{\gamma\alpha}c_\alpha$, where $W_{\gamma\alpha}$ are just real numbers, meaning they commute

$$\{b_\gamma, b_\delta\} = \sum_{\alpha,\beta} W_{\gamma\alpha}W_{\delta\beta}\{c_\alpha, c_\beta\} \tag{2.17}$$

$$= \sum_{\alpha,\beta} W_{\gamma\alpha}W_{\delta\beta}2\delta_{\alpha\beta} \tag{2.18}$$

$$= 2\sum_{\alpha} W_{\gamma\alpha}W_{\delta\alpha} \tag{2.19}$$

$$= 2\sum_{\alpha} W_{\gamma\alpha}W^T_{\alpha\delta} \tag{2.20}$$

$$= 2\delta_{\gamma\delta} \tag{2.21}$$

where the last step is a result of W being orthogonal. The second property stems from the antisymmetry of A, such that in this basis A acquires a canonical block-diagonal form

$$WAW^T = \begin{pmatrix} 0 & \epsilon_1 & & & \\ -\epsilon_1 & 0 & & & \\ & & \ddots & & \\ & & & 0 & \epsilon_N \\ & & & -\epsilon_N & 0 \end{pmatrix}. \tag{2.22}$$

This condition stems from a mathematical theorem on the canonical form of real antisymmetric matrices, which states that if we have a real antisymmetric matrix A of size $2N \times 2N$ there exists a real orthogonal matrix W such that A can be transformed into a block-diagonal form

$$WAW^T = \mathrm{diag}(C_1, C_2, \cdots, C_N), \quad C_j = \begin{pmatrix} 0 & -\lambda_j \\ \lambda_j & 0 \end{pmatrix} \tag{2.23}$$

with $\lambda_j > 0$ being real numbers, stemming from the pairs of complex eigenvalues $\pm i\lambda_j$ for A. This theorem is a standard result in matrix analysis, and a proof can be found in [2] [1]. Once we have a matrix W that satisfies the above conditions we can reduce the Hamiltonian further to a canonical form by inverting the transformation $b_\gamma = \sum_\alpha W_{\gamma\alpha} c_\alpha$ to $c_\alpha = \sum_\gamma W^T_{\alpha\gamma} b_\gamma = \sum_\gamma W_{\gamma\alpha} b_\gamma$

$$H_{\text{canonical}} = \frac{i}{4} \sum_{\alpha,\beta} A_{\alpha\beta} c_\alpha c_\beta \tag{2.24}$$

$$= \frac{i}{4} \sum_{\alpha,\beta,\gamma,\delta} A_{\alpha\beta} W_{\gamma\alpha} b_\gamma W_{\delta\beta} b_\delta \tag{2.25}$$

$$= \frac{i}{4} \sum_{\alpha,\beta,\gamma,\delta} W_{\gamma\alpha} A_{\alpha\beta} W_{\delta\beta} b_\gamma b_\delta \tag{2.26}$$

$$= \frac{i}{4} \sum_{\gamma,\delta} \left(W A W^T \right)_{\gamma\delta} b_\gamma b_\delta \tag{2.27}$$

$$= \frac{i}{4} \sum_{m=1}^{N} \begin{pmatrix} b'_m & b''_m \end{pmatrix} \begin{pmatrix} 0 & \epsilon_m \\ -\epsilon_m & 0 \end{pmatrix} \begin{pmatrix} b'_m \\ b''_m \end{pmatrix} \tag{2.28}$$

$$= \frac{i}{4} \sum_{m=1}^{N} \epsilon_m \left(b'_m b''_m - b''_m b'_m \right) \tag{2.29}$$

$$= \frac{i}{4} \sum_{m=1}^{N} \epsilon_m 2 b'_m b''_m \tag{2.30}$$

$$= \frac{i}{2} \sum_{m=1}^{N} \epsilon_m b'_m b''_m = \sum_{m=1}^{N} \epsilon_m \left(\tilde{a}^\dagger_m \tilde{a}_m - \frac{1}{2} \right) \quad (\epsilon_m \geq 0). \tag{2.31}$$

Where we used that $\{b'_m, b''_m\} = 0$ as they are separate components of b. In this form b'_m and b''_m are real linear combinations of c_{2j-1} and c_{2j}, retaining the commutation relations. N is the number of fermionic sites in the system, here $N = L$. The complex fermion quasiparticles are defined as $\tilde{a}_m = \frac{1}{2}\left(b'_m + i b''_m\right)$ and $\tilde{a}^\dagger_m = \frac{1}{2}\left(b'_m - i b''_m\right)$.

The key take-away here is that the Kitaev chain Hamiltonian, due to being quadratic, can be diagonalized into independent quasi particles $\tilde{a}_m$ with energies ϵ_m. This is how the problem is reduced from a 2^L-dimensional many-body problem to single-particle energies and modes on a $2L \times 2L$ matrix.

2.2 Topological phases

The bulk spectrum for energy vs. momentum is given by

$$\epsilon(q) = \pm \sqrt{\left(2\omega \cos(q) + \mu\right)^2 + 4|\Delta|^2 \sin^2(q)}, \quad -\pi \leq q \leq \pi. \tag{2.32}$$

Finding this relation is done by Fourier transforming the Hamiltonian to momentum space and finding its eigenvalues as shown in appendix A.2.

From this we get the plot of topological values in Eivinds(?)

2.3 BdG

[1] There are apparently many books on this topic, none of which i had access to. This is the best source i could find without proving it directly.

Chapter 3

Quantum computation

The main idea for using Majorana fermions for quantum computation is through a process called braiding. In this process the Majoranas are spatially rotated around eachother, producing a difference in phase. In 1D this is however hard to do; rotating the whole chain around itself has no effect. And if two Majoranas meet, they simply combine into a normal fermion, allowing the stored information to decohere. Several methods are being discussed to overcome this obstacle, and we will study two of them.

3.1 Logical qubits from Majorana modes

3.1.1 Degenerate ground states

Recalling the previously discussed Kitaev chain Hamiltonian, and the following definition using the Majorana operators c_m with properties

$$d_j = \frac{1}{2}\left(c_{2j} + ic_{2j+1}\right) \tag{3.1}$$

$$d_j^\dagger = \frac{1}{2}\left(c_{2j} - ic_{2j+1}\right) \tag{3.2}$$

$$c_m^\dagger = c_m, \quad \{c_l, c_m\} = 2\delta_{lm}. \tag{3.3}$$

We recall that in the trivial phase, Majoranas on all sites pair up into ordinary fermions with non-zero energy. However in the topological phase the chain experiences unpaired Majoranas at the edges, which we want well-separated (meaning their wavefunctions overlap negligibly). For a 1D wire of $2M$ such edge Majoranas one can construct M fermion operators as

$$f_j = \frac{1}{2}\left(c_{2j} + ic_{2j+1}\right). \tag{3.4}$$

These correspond to zero-energy modes, and can give rise to 2^M degenerate ground states which can be labeled by occupation numbers $n_j = f_j^\dagger f_j = 0, 1$.

Wave function representation

Conservation of parity

Four Majoranas - a proper qubit

Nonlocality - spatial separation of Majoranas

[3] [4]

Figure 3.1

3.2 Braiding and non-abelian statistics (exchange of Majoranas)

3.2.1 General result

Berry phase

3.2.2 Clifford group

?

3.2.3 What's missing

Something something non-topological
supplementary operations

3.3 Methods using the Kitaev chain

3.3.1 T-junction

The T-junction is a proposed architecture for allowing braiding with a 1D Kitaev chain. Simply put, the idea stems from connecting a 1D chain to the middle of another, producing a T-shape. In this geometry, one can in theory braid the Majoranas by altering the parameters of the Hamiltonian to rotate the edge states around eachother without touching.

Exchange of two Majoranas

Network of wires

3.3.2 Quantum dots

3.3.3 Others

Chapter 5

Spørsmål

Generelle spørsmål: 08.03.26: Bruker jeg bra notasjon eller burde jeg bytte til mer kompakt notasjon (dot for derivert)?

Eivind master:

Kitaev original article https://arxiv.org/pdf/cond-mat/0010440 - Hvordan kommer majorana fermionene til kantene - er det bare et matematisk triks ved å definere hamiltonianen ved riktige operatorer? Ref. figur i Kitaev artikkel

"Introduction to topological superconductivity and Majorana fermions Martin Leijnse and Karsten Flensberg " https://arxiv.org/pdf/1206.1736

Non-Abelian statistics and topological quantum information processing in 1D wire networksfile:///C:/Users/axlkl/Downloads/nphys1915.pdf

From its supplementary: -

Topological Quantum Computation https://arxiv.org/pdf/quant-ph/0101025

5.1 Huskeliste / idemyldring

Recap av Kitaev artikkelen om Majorana fermioner - Ikke bruk Arxiv som kilde - Flesh ut Kitaev kapittel med - Hvorfor operatorene er som de er mtp. hermitisk, antikommuterende - Mer kondenserte språk? - Mer direkte eksempler på hva Majoranaene tåler - Matematiske nøyaktigheter og innsikter

Adiabatisk (betydning)

Kutte kjede - Hvordan kjeden er kuttet; hva som faktisk endres og hva det tilsvarer fysisk

T-junction - Flytte rundt på fermionene. Hvordan de flyttes rundt, helt spesifikt - Hva skjer når to ender møtes og hele kjeden blir topologisk? Forsvinner noen majoranaer? Er dette lov? - Altså 3 ender - Trikant form vs. Perfekt - Josephson current fra fasen til superledende ledd kan gi problemer?

Sammenligne med Eivind - god kode - Resultater for kutte kjede - Diskusjon mtp. RK4 vs. euler cromer; nøyaktighet, tidssteg, tidsbruk, trunkering osv.

Gjøre: - Lese quantum dot Vs T junction artikkel frem til kap. B

Bibliography

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[2] Howard E. Haber. *Properties of Antisymmetric Matrices*. 2015. URL: https://scipp-legacy.pbsci.ucsc.edu/~haber/ph218/pfaffian15.pdf (visited on 19/08/2026).

[3] Jason Alicea et al. 'Non-Abelian Statistics and Topological Quantum Information Processing in 1D Wire Networks'. In: *Nature Physics* 7.5 (May 2011), pp. 412–417. ISSN: 1745-2473, 1745-2481. DOI: 10.1038/nphys1915. URL: https://www.nature.com/articles/nphys1915 (visited on 29/01/2026).

[4] Jason Alicea. 'New Directions in the Pursuit of Majorana Fermions in Solid State Systems'. In: *Reports on Progress in Physics* 75.7 (1st July 2012), p. 076501. ISSN: 0034-4885, 1361-6633. DOI: 10.1088/0034-4885/75/7/076501. arXiv: 1202.1293 [cond-mat.supr-con]. URL: http://arxiv.org/abs/1202.1293 (visited on 24/08/2026).